

Manish Barti

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SUMMARY

Final year Computer Science undergraduate with hands-on experience building software projects, solving problems, and working with modern technologies

EDUCATION

Graphic Era Deemed to be University

Dehradun, India

B.Tech, Computer Science Engineering

2023 – 2027

PROJECTS

SafeNav - Web & Link Safety Analysis Platform | *Redis, asyncio, PostgreSQL* [Demo](#) | [GitHub](#)

- Built SafeNav, a web and link safety platform that analyzes URLs using lexical, SSL/TLS, IP intelligence, content, and ML-based security signals.
- Designed a multi-layer threat pipeline with compound risk scoring, providing auditable per-rule risk breakdowns instead of black-box verdicts.
- Reduced scan latency to **under 3s** by parallelizing 5 network modules using `asyncio.gather` with Redis-backed caching.
- Trained a Random Forest classifier on **650K+** labeled URLs using 17 lexical features and built a **100K-domain** in-memory trust registry to improve detection and reduce false positives.

GigInsurance - Parametric Income Insurance Platform

[GitHub](#)

- Built GigInsurance, an end-to-end parametric income insurance platform for gig delivery workers, combining AI-driven claims, RAG, WhatsApp integration, and automated payouts.
- Designed a LangGraph-based autonomous claim agent with 6 real-time verification tools, including weather APIs, government feeds, and policy RAG, to automate claim verification.
- Reduced claim processing time to **under 30 seconds** by using asynchronous FastAPI services and parallel real-time verification.
- Implemented XGBoost risk scoring, Isolation Forest fraud detection, parametric triggers, and ChromaDB-based RAG to automate risk assessment and claim decisions.

APIGate - Distributed API Gateway | *FastAPI, Redis, Docker, Grafana*

[GitHub](#)

- Built APIGate, a distributed API Gateway for routing and managing requests across multiple containerized backend services.
- Designed the gateway with Redis-backed sliding-window rate limiting, JWT authentication, request routing, and a circuit breaker for service reliability.
- Optimized request handling to support **20K+ requests/min** while monitoring p99 latency, throughput, and error rates using Prometheus and Grafana.
- Containerized the complete microservices stack using Docker Compose, enabling **one-command deployment** of the gateway, Redis, monitoring, and mock backend services.

TECHNICAL SKILLS

Programming:	C/C++, Python, SQL
Web Development:	HTML, CSS, JavaScript, React.js
Backend :	FastAPI, Express.js, REST APIs, JWT Auth
Databases:	PostgreSQL, MySQL, Redis, Vector DB
Generative AI/ML:	LangChain, LangGraph, RAG, NLP, XGBoost, Isolation Forest, Pandas, NumPy
Cloud & DevOps:	AWS EC2, AWS S3, Azure, Docker, Linux, Prometheus, Grafana, Kubernetes
Tools:	GitHub, Postman, VS Code, PyCharm

CERTIFICATIONS

AWS Cloud Essentials AWS Skill Builder

Jan – June 2026

AWS AI Practitioner AWS Skill Builder

Jul – Nov 2025

LEADERSHIP & INVOLVEMENT

Member, Google Developer Student Clubs (GDSC)

2023 – 2024